Complex Numbers: Cartesian Form & Algebra

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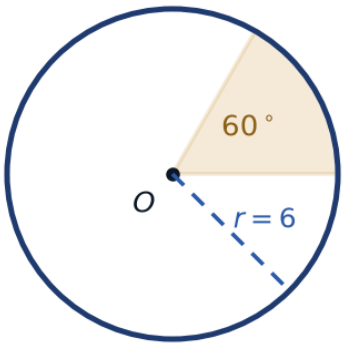
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Cartesian Arithmetic & Complex Conjugate**

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

2 **[Foundational]** • **Target: Modulus and Argument**

Express $z = -1 + i\sqrt{3}$ in polar form $r(\cos\theta + i\sin\theta)$ and exponential form $re^{i\theta}$, with principal argument $-\pi < \theta \leq \pi$. Hence find z^6 and determine all three solutions to $w^3 = z$.



3 **[Foundational]** • **Target: Solving Complex Equations**

Consider the complex number w such that $w = x + yi$, where x and y are real numbers. Given that $w^3 = -64i$ and the imaginary part of w is positive, find the value of $x^2 + y^2$.

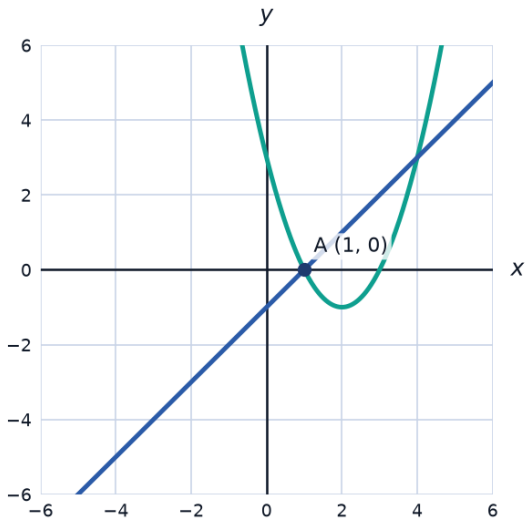
4 **[Foundational]** • **Target: Argand Diagram Geometry & Loci**

Let z_1 and z_2 be complex numbers such that $z_1 = 2 + i$ and $z_2 = 3 - 2i$. ; Consider the complex number $w = \frac{z_1^2}{z_2}$. ; If $w = a + bi$, where a and b are real numbers, find the value of $a^2 + b^2$.

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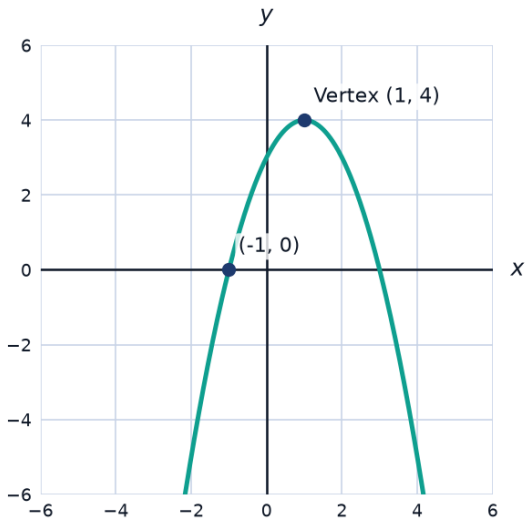
5 [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

The diagram shows the graph of the function $f(x) = x^2 - 4x + 3$ and a line $y = g(x)$. The line $y = g(x)$ intersects the parabola $y = f(x)$ at points A and B. Point A is given as $(1, 0)$. Find the x-coordinate of point B.



6 [Foundational] • Target: Modulus and Argument

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(1, 4)$, and it passes through the point $(-1, 0)$. Find the value of b.



7 [Foundational] • Target: Solving Complex Equations

For the rational function $f(x) = \frac{x^2 - x - 2}{x - 3}$: (a) find the equations of all asymptotes; (b) find the coordinates of any stationary points and determine their nature.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Argand Diagram Geometry & Loci**

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \leq \pi$. Show that the sum of these five roots equals 0.

9 **[Intermediate]** • **Target: Cartesian Arithmetic & Complex Conjugate**

The diagram shows two similar triangles ABC and ADE, where D lies on AB and E lies on AC. It is given that AD = 6 cm, DB = 4 cm, and DE = 12 cm. Find the length of BC.

10 **[Intermediate]** • **Target: Modulus and Argument**

Find the Cartesian symmetric equation of the line of intersection between planes $\pi_1 : x + y + z = 1$ and $\pi_2 : 2x - y + 3z = 4$.

11 **[Intermediate]** • **Target: Solving Complex Equations**

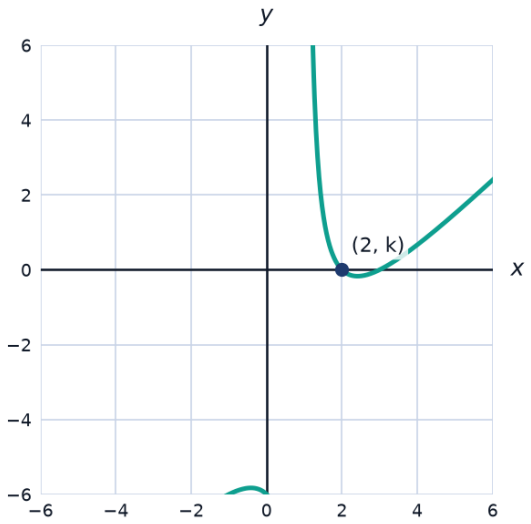
Consider the complex number z such that $z = a + 5i$ and $|z| = \sqrt{26}$. Given that a is a real number, find the possible values of a.

12 **[Intermediate]** • **Target: Argand Diagram Geometry & Loci**

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

13 **[Intermediate]** • **Target: Cartesian Arithmetic & Complex Conjugate**

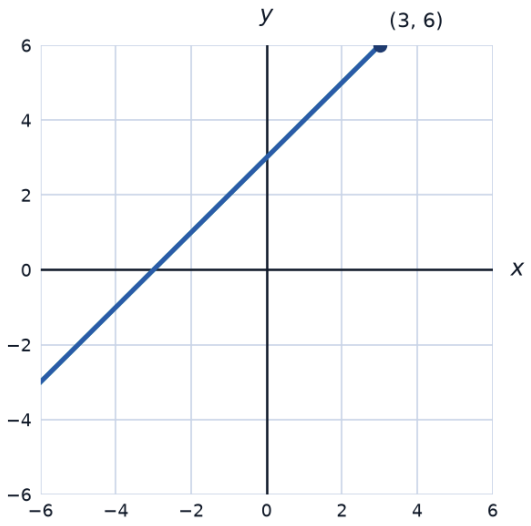
The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.



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14 [Intermediate] • Target: Modulus and Argument

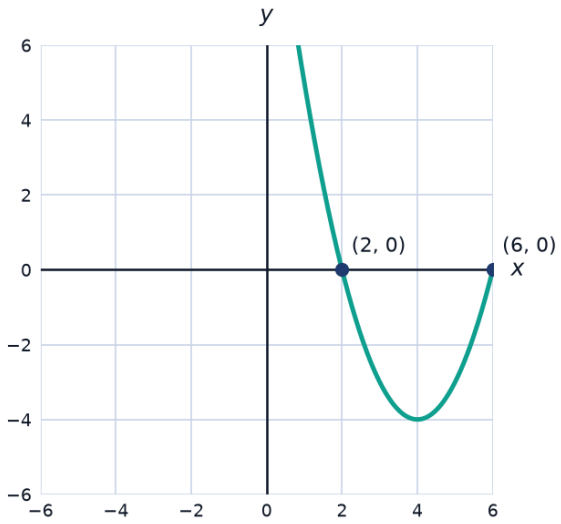
Consider the function $f(x) = \frac{x^2 - 9}{x - k}$. The graph of $y = f(x)$ has a removable discontinuity at $x = 3$. Find the value of k. The graph of $y = f(x)$ is shown below.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

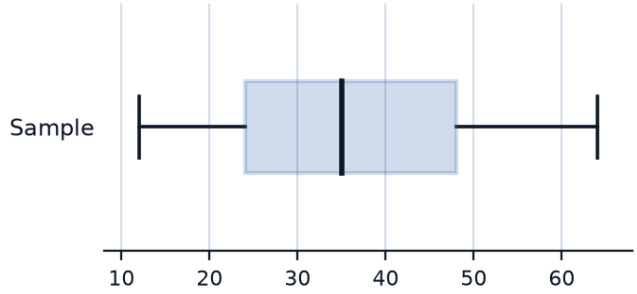
Consider the quadratic function $f(x) = x^2 - 8x + k$. The graph of $y = f(x)$ is a parabola that intersects the x-axis at the points $(2, 0)$ and $(6, 0)$. Find the value of k.



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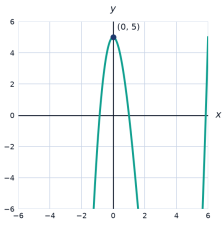
16 **[Challenging]** • **Target: Argand Diagram Geometry & Loci** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\mathrm{Var}(X)$.



17 **[Challenging]** • **Target: Cartesian Arithmetic & Complex Conjugate** • **Cross-Lesson: 1.01 Proof by Mathematical Induction**

The graph of the function $f(x) = x^3 - 6x^2 + 5$ is shown below. The graph has a local maximum and a local minimum. Find the x-coordinate of the local minimum.

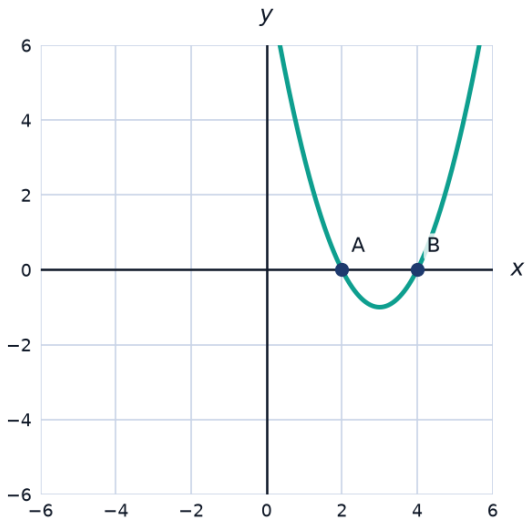


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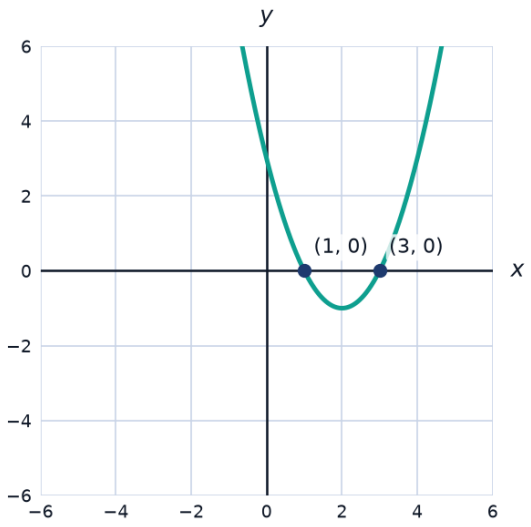
18 [Challenging] • Target: Modulus and Argument • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A?



19 [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 1.04 Partial Fractions

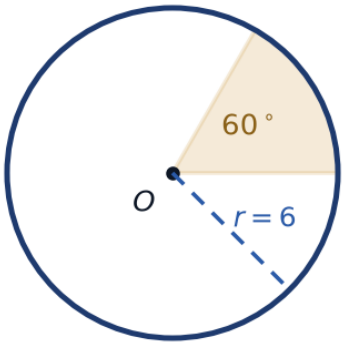
The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. Determine the sum of the x-intercepts.



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20 [Challenging] • Target: Argand Diagram Geometry & Loci • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, calculate $z_1 z_2$.

a. $5 + 14i$

b. $-3 - 8i$

c. $5 - 14i$

d. $-3 + 14i$

22 [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

Express $\frac{4+2i}{1-i}$ in Cartesian form $a + bi$.

a. $1 + 3i$

b. $3 + 3i$

c. $1 - 3i$

d. $2 + 2i$

23 [Foundational] • Target: Cartesian Arithmetic & Complex Conjugate

If $z = x + iy$, what is $z \cdot z^*$ (where z^* is the complex conjugate)?

a. $x^2 + y^2$

b. $x^2 - y^2$

c. $x^2 + 2ixy - y^2$

d. $2x$

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24 [Foundational] • Target: Modulus and Argument

Find the modulus $|z|$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$ for $z = -\sqrt{3} + i$.

a. $|z| = 2, \text{Arg}(z) = \frac{5\pi}{6}$

b. $|z| = 2, \text{Arg}(z) = -\frac{\pi}{6}$

c. $|z| = 4, \text{Arg}(z) = \frac{2\pi}{3}$

d. $|z| = 2, \text{Arg}(z) = \frac{7\pi}{6}$

25 [Foundational] • Target: Modulus and Argument

What is the modulus of $\frac{3+4i}{1-2i}$?

a. $\sqrt{5}$

b. $\frac{5}{3}$

c. 5

d. $\frac{25}{5} = 5$

26 [Foundational] • Target: Modulus and Argument

Find the principal argument of $z = -5i$.

a. $-\frac{\pi}{2}$

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{2}$

d. π

27 [Foundational] • Target: Solving Complex Equations

Solve the equation $2z + 3z^* = 15 - 2i$ for $z = x + iy$.

a. $z = 3 + 2i$

b. $z = 3 - 2i$

c. $z = 5 - 2i$

d. $z = 2 + 3i$

28 [Foundational] • Target: Solving Complex Equations

Find the two square roots of $-8 - 6i$ in Cartesian form.

a. $\pm(1 - 3i)$

b. $\pm(3 - i)$

c. $\pm(1 + 3i)$

d. $\pm(2 - 3i)$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Solving Complex Equations**

Solve the quadratic equation $z^2 - 4z + 13 = 0$ over $\mathbb{C}$.

a. $z = 2 \pm 3i$

b. $z = 4 \pm 6i$

c. $z = -2 \pm 3i$

d. $z = 2 \pm 9i$

30 **[Intermediate]** • **Target: Argand Diagram Geometry & Loci**

Describe the locus of points in the complex plane satisfying $|z - 3 + 2i| = 5$.

a. A circle with center $(3, -2)$ and radius 5

b. A circle with center $(-3, 2)$ and radius 5

c. A straight line perpendicular to $3 - 2i$

d. An ellipse with focal distance 5

31 **[Intermediate]** • **Target: Argand Diagram Geometry & Loci**

What geometric locus is defined by $|z - 1| = |z - 3i|$?

a. The perpendicular bisector of the segment connecting $(1, 0)$ and $(0, 3)$

b. A circle passing through $(1, 0)$ and $(0, 3)$

c. A line segment connecting $(1, 0)$ and $(0, 3)$

d. A parabola opening upward

32 **[Intermediate]** • **Target: Argand Diagram Geometry & Loci**

Find the Cartesian equation of the locus $\text{Arg}(z - 2) = \frac{\pi}{4}$.

a. The half-line $y = x - 2$ for $x > 2$

b. The full line $y = x - 2$

c. The circle $(x - 2)^2 + y^2 = 1$

d. The half-line $y = -x + 2$ for $x < 2$

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33 [Intermediate] • Target: Cartesian Arithmetic & Complex Conjugate

How many distinct complex nth roots does the equation $z^7 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 7 roots, $\frac{360}{7}^\circ$ apart

b. 8 roots, $\frac{360}{7}^\circ$ apart

c. 7 roots, 45° apart

d. 6 roots, $\frac{360}{7}^\circ$ apart

34 [Intermediate] • Target: Modulus and Argument

How many distinct complex nth roots does the equation $z^3 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 3 roots, 90° apart

b. 2 roots, 120° apart

c. 3 roots, 120° apart

d. 4 roots, 120° apart

35 [Intermediate] • Target: Solving Complex Equations

How many distinct complex nth roots does the equation $z^5 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 5 roots, 72° apart

b. 4 roots, 72° apart

c. 5 roots, 60° apart

d. 6 roots, 72° apart

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36 [Intermediate] • Target: Argand Diagram Geometry & Loci

How many distinct complex nth roots does the equation $z^4 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 4 roots, 72° apart

b. 4 roots, 90° apart

c. 5 roots, 90° apart

d. 3 roots, 90° apart

37 [Intermediate] • Target: Cartesian Arithmetic & Complex Conjugate

How many distinct complex nth roots does the equation $z^8 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 9 roots, 45° apart

b. 8 roots, 40° apart

c. 7 roots, 45° apart

d. 8 roots, 45° apart

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Modulus and Argument • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Consider the complex number $w = 4 - 3i$. Find the modulus of w, denoted as $|w|$.

a. 5

b. 6

c. 7

d. 25

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39 [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 1.04 Partial Fractions

Let $z = x + yi$ be a complex number. The modulus of z, denoted by $|z|$, is given by $|z| = \sqrt{x^2 + y^2}$. Given that $|z| = \sqrt{5}$ and the real part of z is 2, find the possible values for the imaginary part of z.

a. $\pm\sqrt{3}$

b. ± 3

c. $\pm\sqrt{1}$

d. ± 1

40 [Challenging] • Target: Argand Diagram Geometry & Loci • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. 5

b. 7

c. 25

d. $\sqrt{7}$

41 [Challenging] • Target: Cartesian Arithmetic & Complex Conjugate • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Consider the complex number $z = 3 + ki$, where k is a real parameter. The modulus of z is given by $|z| = \sqrt{13}$. Find the positive value of k.

a. 1

b. 2

c. 3

d. 4

42 [Challenging] • Target: Modulus and Argument • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Consider the complex number $z = -3 + 4i$. Find the modulus of z, denoted by $|z|$.

a. 5

b. 7

c. 25

d. 16

43 [Challenging] • Target: Solving Complex Equations • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Write $z = 5(\cos 135^\circ + i\sin 135^\circ)$ in the form $x + yi$ (exact values).

a. $-\frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

b. $\frac{5\sqrt{2}}{2} - \frac{5\sqrt{2}}{2}i$

c. $\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$

d. $-\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$

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44 [Challenging] • Target: Argand Diagram Geometry & Loci • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Write $z = 3(\cos 135^\circ + i\sin 135^\circ)$ in the form $x + yi$ (exact values).

a. $-\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$

b. $-\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$

c. $\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$

d. $\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$

45 [Challenging] • Target: Cartesian Arithmetic & Complex Conjugate • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Write $z = 3(\cos 150^\circ + i\sin 150^\circ)$ in the form $x + yi$ (exact values).

a. $-\frac{3\sqrt{3}}{2} + \frac{3}{2}i$

b. $\frac{3}{2} - \frac{3\sqrt{3}}{2}i$

c. $\frac{3\sqrt{3}}{2} + \frac{3}{2}i$

d. $-\frac{3\sqrt{3}}{2} - \frac{3}{2}i$

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